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Use of liquid chromatography quadrupole time-of-flight mass spectrometry and metabolomic approach to discriminate coffee brewed by different methods

Lei Xu[a,b], Fei Lao[b], Zhenzhen Xu[a,*], Xue Wang[a], Fang Chen[b], Xiaojun Liao[b], Ailiang Chen[a], Shuming Yang[a]

[a] Institute of Quality Standard & Testing Technology for Agro-Products, Chinese Academy of Agricultural Sciences, Key Laboratory of Agro-food Safety and Quality, Ministry of Agriculture, Beijing 100081, China

[b] Beijing Advanced Innovation Center for Food Nutrition and Human Health, College of Food Science and Nutritional Engineering, China Agricultural University, Beijing Key Laboratory for Food Nonthermal Processing, Key Lab of Fruit and Vegetable Processing, Ministry of Agriculture, Beijing 100083, China

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ABSTRACT

An untargeted metabolomic method based on UPLC-QTOF were used to investigate the differences in coffee brewed by boiled, pour-over and cold-brew methods here. Distinctive separation among the three groups could be seen from principal component analysis and hierarchical clustering analysis. Analysis of variance, fold change and orthogonal projection to latent structures discriminant mode were conducted to find the characteristic potential markers, subsequently, nine potential markers were putatively identified using general chemical databases, and five of them were further confirmed by acquisition of reference standards. This work provides an efficient way for discrimination of coffee brewed by different methods. Interestingly, the result of this work also suggested that the contents of two selected markers, norharman and harman, were higher in the pour-over and boiled methods, compared to the cold-brew method. This content difference were further verified by the quantitative analysis data of commercial coffee samples.

1. Introduction

Coffee (*Coffea* sp.) plant is widely cultivated in the world, and *Coffea arabica* and *Coffea canephora variant robusta* are the most popular. They account for approximately 75% and 24% of global coffee production (Choi, Choi, Park, Lim & Kwon, 2010). Brewed coffee is consumed by millions of people every day (Esquivel & Jiménez, 2012). During the past few years, evidence of its health benefits, such as the contribution of coffee to the intake of antioxidants in the diet, has helped to increase coffee consumption (Perez-Martinez, Caemmerer, De Pena, Cid & Kroh, 2010).

Coffee is brewed in many ways depending on consumer preference (Moeenfard, Rocha & Alves, 2014). Consumer's choice for particular coffee varies by numerous parameters (Moeenfard et al., 2014). Brew method is one of these critical parameters, as it may influence the extraction rates of many chemical species identified in roasted coffee (Petracco, 2005). In filtered coffee, a U-shape extraction profile was observed for antioxidants over the prolonging of brew time. Thanks to the turbulences generated in the filter coffeemaker, less polar antioxidant compounds extraction efficiency increased as the longer extraction time (Ludwig, Sanchez, Caemmerer, Kroh, De Peña & Cid, 2012). Mocha brew had the highest yield in coffee antioxidants extraction per gram of ground roasted coffee, compared with espresso, plunger, and filter, while the espresso was richest in terms of antioxidant intake (per milliliter of coffee brew) (Perez-Martinez et al., 2010). Caffeoylquinic acids were detected in high levels in light roasted boiled brews, while lower levels were detected in dark roasted filtered brews (Tfouni, Carreiro, Teles, Furlani, Cipolli & Camargo, 2014). Noted that the degree of roasting also plays a role in caffeoylquinic acid content (Tfouni et al., 2014).

In addition, cold-brew, a method minimizes thermal processing, is becoming more and more popular. Roast Magazine reported a 460% increase in retail sales of refrigerated cold brew coffee in the United States from 2015 to 2017, generating $38 million in 2017 alone (https://dailycoffeenews.com/2017/10/03/new-us-coffee-shop-growth-slows-as-rtd-and-cold-brew-accelerate-according-to-mintel. Accessed 13.12.19). Cold-brew utilizes lukewarm to refrigerated water and steeps the coffee grinds of 12 h (Lane, Palmer, Christie, Ehlting &

* Corresponding author.
E-mail address: xuzhenzhen@caas.cn (Z. Xu).

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Le, 2017). It was reported that cold brew coffee was found to have lower total titratable acid and antioxidant activity than their hot brew counterparts (Rao & Fuller, 2018). When comparing cold brew and cold drip, cold drip coffee was perceived as more bitter, due to greater contents of caffeine and chlorogenic compounds than cold brews (Angeloni, Guerrini, Masella, Innocenti, Bellumori & Parenti, 2018). Application of ultrasonication and agitation in cold brew coffee would lead to significant increases in the color values, total soluble solids, antioxidant activities, chlorogenic acid and caffeine contents (Ahmed, Jiang, Park, Lee, Seok & Eun, 2018).

A wide range of technologies have been used for the authentication of foods, such as isotope ratio mass spectrometry (IRMS) (Camin et al., 2017), liquid chromatography mass spectrometry (LC-MS) (Sidwick, Johnson, Adam, Pereira & Thompson, 2017), gas chromatography mass spectrometry (GC-MS) (Hansen, Frandsen & Fromberg, 2016), near infrared spectroscopy (NIRS) (Ellis, Ellis, Muhamadali, Xu, Horn & Goodacre, 2016) and nuclear magnetic resonance spectroscopy (NMR) (Cicero et al., 2015). As for coffee, NMR fingerprinting profile analysis was used to discriminate the geographical origin of coffee from different countries and continents (Arana et al., 2015). LC-MS was used to identify phenolic compounds in coffee, producing distinct profiles of coffee beans from major producing regions of Ethiopia (Mehari, Redi-Abshiro, Chandravanshi, Combrinck, Atlabachew & McCrindle, 2016).

Metabolomics, the downstream discipline of genomics and proteomics, is defined as the full complement of small molecule metabolites found in a specific cell, organ, or organism (Scalbert et al., 2009). Metabolomics has been utilized in food industry for the detection of fraudulent activities, specifically, for the verification of meat and oil products, including the quality and adulteration (Gil-Solsona et al., 2016; Sidwick et al., 2017). Metabolomics has been applied to some aspects of coffee processing. For example, the coffee metabolomic profile and quality was affected cherries processing, roasting, grinding as well as brewing methods (Selmar, Bytof, Knopp & Breitenstein, 2006). Mainly due to germination processes and stress metabolism (Bytof et al., 2007), various metabolic reactions are presented in green coffee beans dry or wet processing (Selmar et al., 2006). Because of these metabolic processes, sugar contents of wet processed coffee beans are lower than dry-processed ones (Kleinwächter & Selmar, 2010). Although coffee species and brewing method were shown to influence the levels of caffeine in coffee brews (Tfouni et al., 2014), there is still, very limited research were published regarding to coffee hot and cold brew method discrimination based on comprehensive metabolomics.

In this study, experiment was conducted to compare the metabolomics of coffee brewed by three methods (pour-over, boiled, and cold-brew). The aim of this study was to discover and identify relevant markers that allow discriminating coffee brewed by different methods using ultra performance liquid chromatography (UPLC) coupled to quadrupole time-of-flight mass spectrometry (QTOF-MS), in combination with multivariate analysis.

2. Materials and methods

2.1. Chemicals and reagents

Acetonitrile was purchased from Thermo Fisher Scientific (Massachusetts, USA), and formic acid was purchased from DiKMA Technologies (Beijing, China). Water was generated by a Milli-Q integral water purification system (Millipore Billerica, MA). The ammonium acetate and standards of norharman, 1-palmitoyllysophosphatidylcholine, 4-hydroxy-3-methoxycinnamaldehyde, harman, pyrimethanil were purchased from the Sigma-Aldrich (St. Louis, USA).

2.2. Coffee samples

To avoid variations in potential markers related to the coffee bean origin, green coffee beans from three representative geographic regions (Asia, South America, and Africa) were selected for analysis. The green coffee beans used in this study were C. arabica L. (arabica) of Yirgacheffe (Ethiopia), Huilia (Colombia), and Simao (YunNan, China). The coffee beans were roasted at 216 °C by a local factory for three different time periods: 9 min 31 s, 10 min 20 s, and 10 min 12 s, respectively. Then, the roasted coffee beans were ground by a bean grinder (EK43, MAHLKONIG Gmbh & Co. KG Hamburg, German), with the granular level set to 5.5. Ground roasted coffee powders were stored at 4 °C until use.

Three different coffee brew methods: cold-brew, pour-over, and boiled method, were carried out for each coffee powder. For the cold-brew coffee, 60 mL of ultrapure water (25 °C) and 2 g coffee powder were mixed, well-shaken before storing at 4 °C for 12 h. For the pour-over coffee, 60 mL of boiling ultrapure water was poured onto the ground coffee powder (2 g) in a coffee filter paper. The filtrate was collected within 2 min and allowed to cool to room temperature for 1 h (Sanz, Czerny, Cid & Schieberle, 2002). For the boiled coffee, 60 mL of boiling ultrapure water was added to 2 g of coffee powder, then the brew was stirred and incubated in a boiling water bath for 5 min. The coffee was allowed to cool to room temperature for 1 h (Górnas, Neunert, Baczyński & Polewski, 2009). Each method was carried out in five replicates for each coffee powder sample.

The 14 commercial coffee drink samples (consisting of 7 thermal processed samples and 7 cold-brew samples) were purchased from local markets and coffee shops for the quantification of the selected markers.

2.3. Untargeted metabolomics analysis

The coffee samples were filtered with a 0.45-μm and 0.22-μm filter membrane. Then, the samples were directly analyzed to determine their metabolomic profiles using UPLC-QTOF (1290 and 6545, Agilent, USA). Chromatographic separation was achieved on a C18 column (Eclipse Plus C18, RRHD 1.8 μm, 2.1*100 mm, Agilent, USA) maintained at 40 °C. The mobile phases A and B for positive ion mode were 0.2% formic acid aqueous solution and acetonitrile, respectively. The mobile phase A for negative ion mode was an aqueous solution of 2 mM ammonium acetate and 0.2% formic acid; mobile phase B was acetonitrile. The flow rate was 0.3 mL/min, and the gradient program was as follows: 0% B to 40% B (in 8 min); increase to 100% B (over 4 min); reduce to 0% B immediately. The post-run time was 2 min. The sample injection volume was 20 μL. Mass spectral data were collected in both the positive and negative ionization mode. The gas temperatures were 320 °C and 250 °C; the sheath gas temperatures were 325 °C and 400 °C; and the capillary voltages were 30 kV and 35 kV for positive mode and negative mode, respectively. Fifteen replicates were run for each sample. A pooled sample (comprising an aliquot of every sample in the study) was injected 5 times of interval to ensure the stability and repeatability of the system.

2.4. Metabolomics data acquisition and processing

The data were acquired and pre-processed (peak detection, data mining, alignment and normalization, removal of isotope masses) using Profinder B.06.00 (Agilent, USA). The following parameters were applied for data acquisition: mass tolerance of 15 ppm, mass window of 0.02 Da, retention time window of 0.25 min, noise elimination level of 300. The absolute height filter was 5000 counts. Data were analyzed using both one-way analysis of variance (ANOVA) and chemometric techniques. Chemometric data analysis was performed using SIMCA 14.1 software (Umetrics, Sweden) and the online tool, metaboanalyst (http://www.metaboanalyst.ca. Accessed 13.06.18). After ANOVA and chemometric analysis, a list of potential markers for the discrimination of the three groups was obtained.

2.5. Identification of potential markers

Target MS/MS mode was used for the MS/MS acquisition of those potential markers. Argon was used as the collision gas, and the collision energy was set as 10 eV, 20 eV and 40 eV, separately. The accurate mass information of precursor was used to calculate the proposed molecular formula of each marker using Qualitative Analysis B.07.00 software (Agilent, USA). Atoms considered for the molecular formula calculation were as follows: C ($3 \le n \le 60$), H ($0 \le n \le 120$), N ($0 \le n \le 30$), O ($0 \le n \le 30$), S ($0 \le n \le 5$) and Cl ($0 \le n \le 3$). The software proposes the formula based on matching of the experimental and theoretical isotopic pattern of the markers. All suggested formulas have very good mass accuracy, less than 4 ppm, thus increasing the confidence in predicted formulas. Then, the molecular formula, exact mass and fragmentation pattern obtained were entered in online chemical databases, metlin and massbank (https://metlin.scripps.edu and http://www.massbank.eu Accessed 06.18) to search for reference candidates.

2.6. Quantification of the selected markers

The selected markers were quantified in 45 experimental samples and 14 commercial drink coffee samples by UPLC coupled with triple quadrupole mass spectrometry (UPLC-QQQ) (1290 and 6470, Agilent, USA), using previous published method with modifications (Truman, Grounds & Brennan, 2017). The UPLC parameters were set as used in untargeted metabolomics analysis. Six different concentration standards of each analyte were run separately (1–50 ppb in acetonitrile), and the external standard method was used. Multiple reaction monitoring mode of mass spectrometric detection was operated in positive ion mode (harman m/z 183.1, norharman m/z 169.1, with subsequent determination of daughter ions at m/z 114.9 and 167.9 for harman, 115.1 and 142.1 for norharman). The injection volume was 5 µL.

3. Results and discussion

3.1. Untargeted metabolomics analysis

The workflow of untargeted metabolomics analysis are shown in Fig. S1. In the present study, 2253 compounds were extracted from the total ion chromatogram of UPLC-QTOF in positive ion mode, and 1071 in negative mode (Figs. S2 and S3). Principal component analysis (PCA) and hierarchical clustering analysis (HCA) were carried out to determine whether the three groups of coffee samples could be differentiated. The PCA scores plots generated for coffee samples brewed by different methods show the separation into distinctive clusters of boiled, pour-over, and cold-brew groups (Fig. 1). The first three principal components (PCs) explain the majority of the variation with PC1, PC2, and PC3 explaining 31.6%, 19.8%, and 14.2% for positive ion mode, and 25.4%, 23.6%, and 17.2% for negative ion mode, respectively.

The compound abundances for the three groups, as determined by HCA, are shown in Fig. 2. The heatmaps display the data with the cases on the horizontal axis and the compounds on the vertical axis. No assumptions are made in HCA, and the correlation between cases is determined independently by the abundance of compound. The clustering of 45 cases in positive ion mode produced 3 groups, consisting of 5 cases per group, can be discerned from the heatmap (Fig. 2A). Group 1, i.e., the cold-brew coffee group, is the first group to branch off in the dendrogram. Groups 2 and 3 are the boiled group and pour-over group, respectively. The lengths of the branches within these three groups are quite short, indicating that the compound profiles of the cases are closely related. Interestingly, each group branched into three subgroups, which happen to be the three different origins of coffee bean. This indicated that the effect of brew methods on coffee compound abundance is greater than that of the growing origin, but the origin of the coffee beans does affect the compound patterns brewed by the same method. In negative ion mode (shown in Fig. 2B), the cluster of 45 samples did not branch into three method groups as in positive ion mode.

3.2. Differential abundance analyses of different coffee samples

To understand the difference in compound abundance between the three brew methods, analysis of variance (ANOVA) and fold change (FC) were used to filter the compounds extracted from raw data. A compound with differential abundance was defined as a compound with p less than 0.05 and FC (compared to the cold-brew group) no less than 2 or no more than 0.5. In this way, 286 and 168 compounds were defined in positive and negative ion modes, respectively.

From the PCA and HCA analyses, more pronounced clustering and significantly better differentiation among the three groups was obtained from positive ion mode. In addition, more differential abundance compounds were detected in positive ion mode. Therefore, only data acquired in positive ion mode were further analyzed in this study.

3.3. Potential markers identification

To find potential characteristic markers for these three groups,

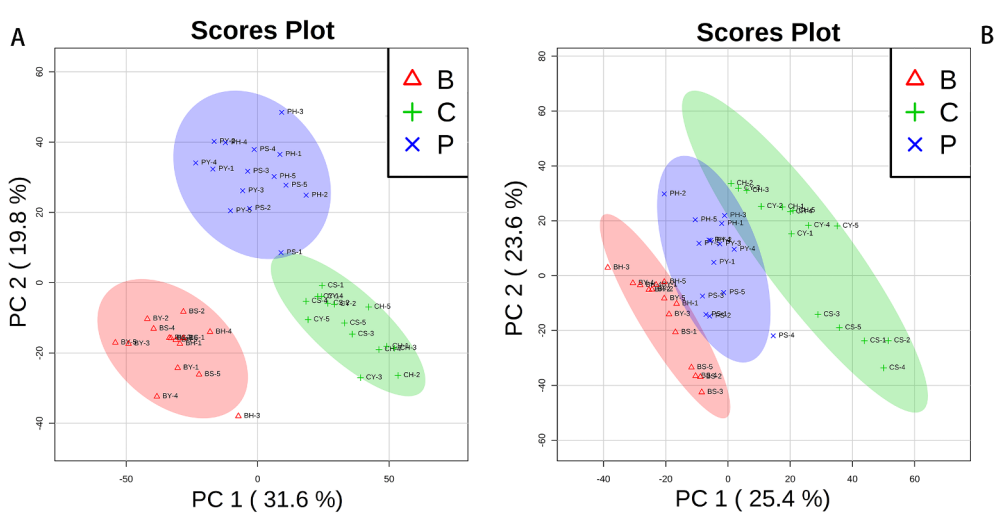

Fig. 1. The PCA scores plots of compound abundances obtained in positive ion mode (A) and negative ion mode (B) from coffee samples brewed by different methods (B, boiled; C, cold-brew; P, pouring over; n = 15; Y, Yirgacheffe from Ethiopia, H, Huilia from Colombia, S, Simao from China).

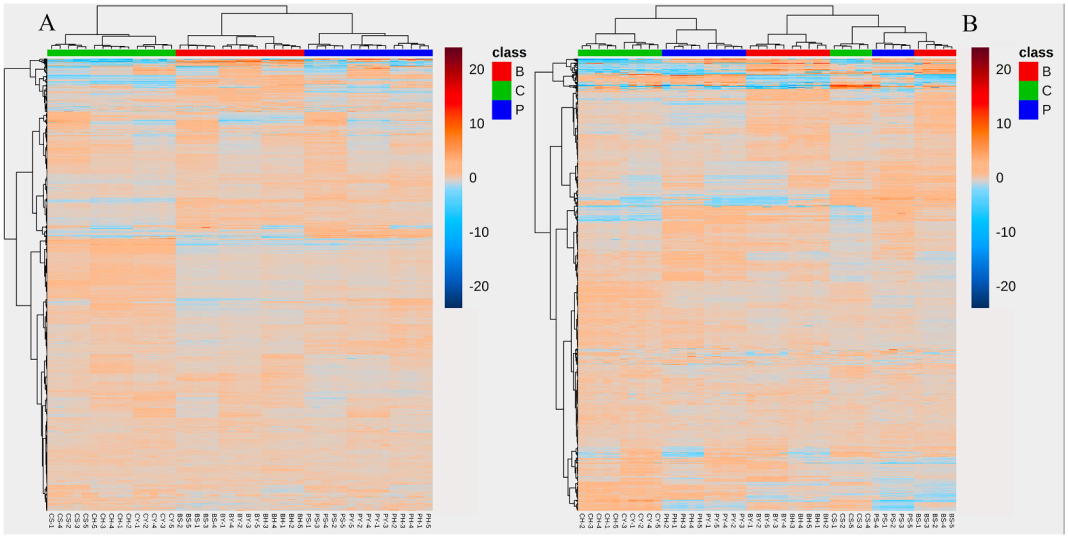

Fig. 2. Heat Maps from hierarchical clustering analysis of compound abundances obtained in positive ion mode (A) and negative ion mode (B) from coffee samples brewed by different methods (B, boiled; C, cold-brew; P, pouring over; n = 15; Y, Yirgacheffe from Ethiopia, H, Huilia from Colombia, S, Simao from China).

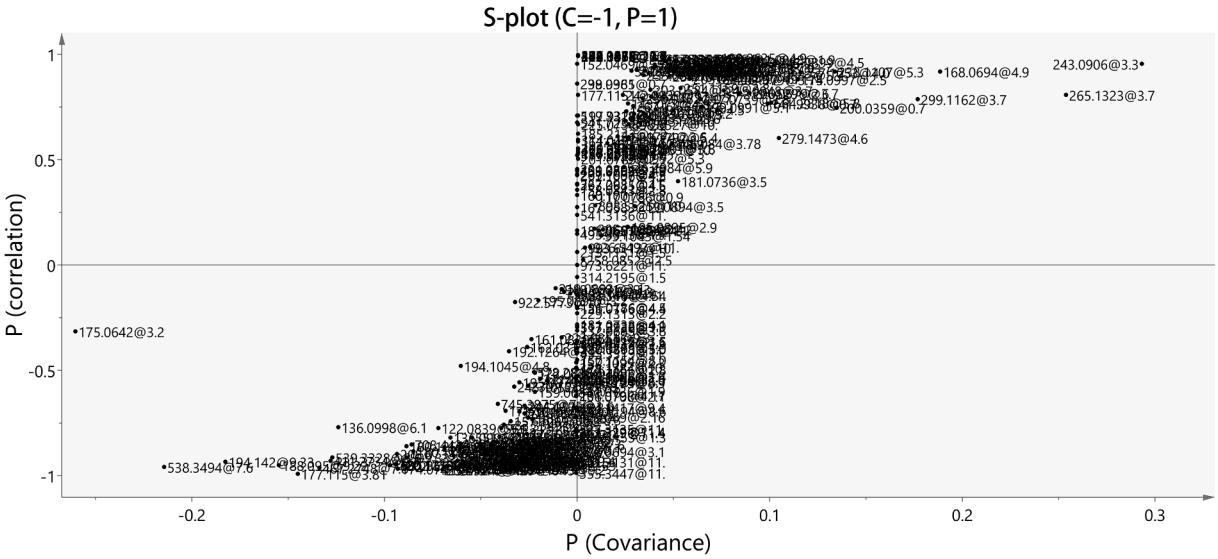

Fig. 3. S-plot of differential compound abundances obtained in positive ion mode from coffee samples brewed by cold-brew (C) and pour-over (P) (n = 15).

orthogonal projection to latent structures discriminant analysis (OPLS-DA) models were built. Potential markers of interest could be selected from the S-plot, constructed from OPLS-DA. Markers were chosen based on their contribution to the variation and correlation within the data set. An example of an S-plot, obtained from OPLS-DA for pour-over coffee and cold-brew coffee, is shown in Fig. 3. Variable importance for projection (VIP) values were generated for all the constructed models to find the variables that contribute the most to classification. In this study, the variables with absolute VIP values greater than 0.9 were selected as potential markers. As shown in Table 1, nine potential markers were putatively annotated by matching with the databases. And the trend plots of nine potential markers, which show the variation in abundance, are shown in Supplementary Material (Fig. S4).

3.4. Characteristic markers confirmation

Next, norharman, harman, pyrimethanil 1-palmitoyllysophosphatidylcholine and 4-hydroxy-3-methoxycinnamaldehyde were unequivocally confirmed by matching the retention times of pure standards using UPLC-QQQ. The five compounds could be used as the characteristic markers to discriminate coffee brewed by boiled, pour-over, or cold-brew methods.

1-palmitoyllysophosphatidylcholine, MS/MS spectrum at 20 eV are shown in Fig. 4A, is a lysophospholipid (Soler-Cantero et al., 2012). It was capable of mitigating dyslipidemia and hyperglycemia by inducing proliferator-activated receptor α (PPARα) activation, and PPARα activation could enhance fatty acid oxidation as well as decrease the levels of circulating and cellular lipids in obese diabetic patients (Haruya et al., 2015). Additionally, as shown in Fig. S4, the peak intensity of Yirgacheffe (Ethiopia) coffee is extremely higher than that of Huilia (Colombia), and Simao (YunNan, China). More information regarding to the contents of 1-palmitoyllysophosphatidylcholine in boiled coffee from different origin should be further studied.

4-hydroxy-3-methoxycinnamaldehyde, MS/MS spectrum at 10 eV are shown in Fig. 4B, a phenolic compound, is found in the heartwood of *Gliricidia sepium* (Herath, Dassanayake, Priyadarshani, Silva, Wannigama & Jamie, 1998). It was reported to have antifungal and cytotoxic activity for various cancer cell lines (Huang-Hui et al., 2011), as well as prevent the melanin production in mouse melanoma cells and primary humanmelanocytes (Roh et al., 2014). Oral Administration (a

Table 1
The identified potential markers between three groups of coffee.

Compound name (confirmed[a] and tentative[b])	RT^1 (min)	RT^2 (min)	Mass	Formula	Collision energy (V)	Matched MS^2 fragments (*m/z*)	VIP value of P/C	VIP value of B/C	VIP value of P/B
Norharman[a]	4.8	4.8	168.0693	C11H8N2	20	142.0629, 115.0533	1.23	1.22	0.19
					40	141.0490, 142.0656, 140.0490, 115.0453, 116.0499, 114.0463			
1,3-Dicaffeoylquinic acid[b]			516.1272	C25H24O12	10	499.1221, 337.0906, 163.0385	1.26	1.24	0.95
					20	499.1235, 319.0827, 163.0392			
					40	163.0387, 145.0280, 135.0438, 117.0331			
1-Palmitoyllysophosphatidylcholine[a]	11.8	11.8	496.3414	C24H51NO7P	10	478.3307, 184.0729, 104.1064	0.59	1.05	1.53
					20	478.3269, 184.0729, 104.1064			
					40	184.0733, 124.9991, 104.1061			
4-Hydroxy-3-methoxycinnamaldehyde[a]	6.8	6.8	178.0630	C10H10O3	10	161.0592, 147.0437	1.29	1.26	1.43
Antipyrine[b]			188.0948	C11H12N2O	20	189.1021, 161.1078, 147.0911, 146.0835	1.25	1.27	1.06
Metalaxyl[b]			279.1459	C15H21NO4	40	149.0595, 145.0277, 134.0348, 121.0747, 106.0398	0.80	1.00	0.78
5-Methoxy-3-indoleacetic acid[b]			205.0743	C11H11NO3	10	206.0805, 16.0762, 146.06, 138.0648, 134.0585	1.25	1.28	0.80
					20	206.0808, 160.0759, 146.0593, 134.06, 118.0654, 106.0648			
Harman[a]	5.2	5.2	182.0847	C12H10N2	10	183.0914, 183.0946, 165.0902	1.16	1.26	0.38
					40	183.091, 168.0678			
Pyrimethanil[a]	8.5	8.5	199.0989	C12H13N3	10	200.1062, 200.071, 182.0613, 105.0694	0.99	1.21	1.01
					20	200.10, 182.0603, 157.0875, 105.0693			
					40	157.0867, 156.0804, 154.0643, 118.0654, 117.0556, 115.0531, 107.0722, 105.0693			

C, cold-brew; P, pour-over; B, boiled.
RT^1 is the retention time of coffee samples, while RT^2 is the retention time of standard samples.
VIP is the variable importance for projection.
[a] Confirmed by standard.
[b] Tentative.

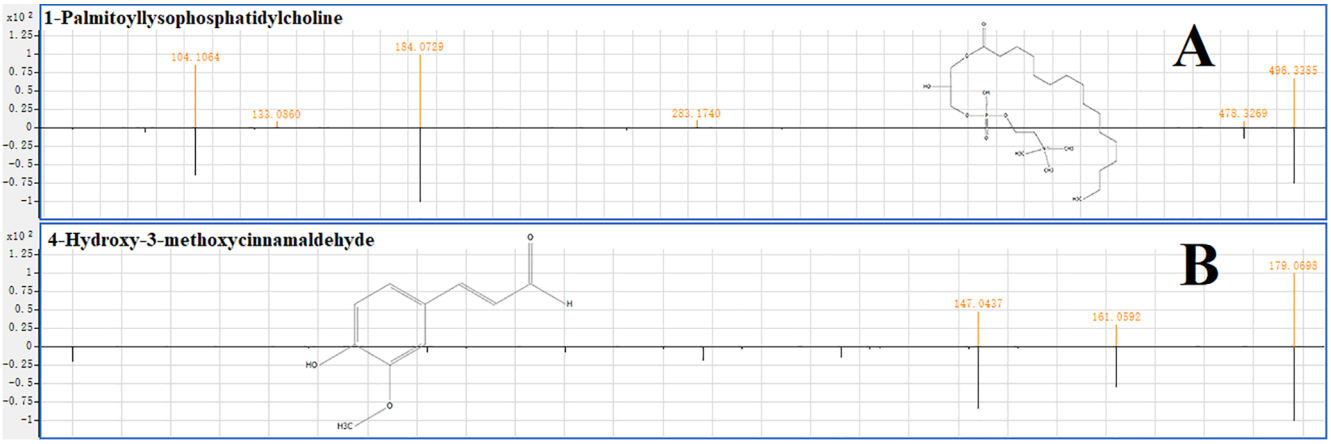

Fig. 4. MS/MS spectrums of 1-palmitoyllysophosphatidylcholine at 20 ev (A) and 4-hydroxy-3-methoxycinnamaldehyde at 10 ev (B) in positive mode.

daily dose was 50 mg/kg, 4 weeks) of 4-hydroxy-3-methoxycinnamaldehyde could also ameliorate atopic dermatitis symptoms by modulating the functions of effector T cells and keratinocytes in mice (Lee et al., 2015).

Pyrimethanil is one of the pesticides (a broad-spectrum fungicide) commonly applied to coffee crops (Dias, Oliveira, Madureira, Silva, Souza & Cardeal, 2013). An increase of pyrimethanil solution in higher temperature significantly improved its concentration in the two thermally processed groups (boiled and pour-over coffee) in this work.

Norharman and harman are two heterocyclic β-carboline alkaloids with biological and potential toxicological activity. They may act as potential mutagens or co-mutagens. Therefore, both of them have been considered as mutagenic and carcinogenic heterocyclic amines (Herraiz, 2004). It was also reported that significantly elevated levels of norharman were observed in patients with Parkinson's disease, both in plasma and in cerebrospinal fluid (Kuhn, Müller, Große, Dierks & Rommelspacher, 1995; Kuhn, Müller, Grosse & Rommelspacher, 1996). On the other hand, harman might also have a critical role in the modulation of learning and memory functions. Therefore, norharman and harman are two compounds with ambiguous functions (Celikyurt, Utkan, Gocmez, Hudson & Aricioglu, 2013). Coffee brew is the major dietary contributor of norharman and harman in humans (Herraiz, 2004; Pfau & Skog, 2004). In brewed coffee (filtered, espresso), 91–165.6 and 26.9–39.9 ng/ml of norharman and harman were detected depending on the strength of the brew (Pfau et al., 2004). Taking into consideration the popularity of coffee, its consumption may have a considerable contribution on norharman and harman intake in the human body (Wojtowicz, Zawirska-Wojtasiak, Przygonski & Mildner-Szkudlarz, 2015).

3.5. Quantification of selected markers

Since pyrimethanil is a cultivating addition, also methodology refinement on the development and validation of a target quantitative methods for 1-palmitoyllysophosphatidylcholine and 4-hydroxy-3-methoxycinnamaldehyde using UPLC-QQQ were still needed, norharman and harman were selected for further verification and quantification in both experimental and commercial coffee samples (Fig. 5). The selected markers, norharman and harman, were quantified in 45 experimental samples, the contents of norharman and harman in the boiled and pour-over groups were higher than in the cold-brew group (Fig. 5). This result was partly in agree with previous report, saying norharman and harman were detected in brewed coffee and in roasted coffee beans, but not in green coffee beans (Wojtowicz et al., 2015). The formation of norharman and harman may be the result of heating and/or pyrolysis of tryptophan, and condensation with aldehydes (Herraiz, 2004). As both compounds were formed in the coffee bean roasting

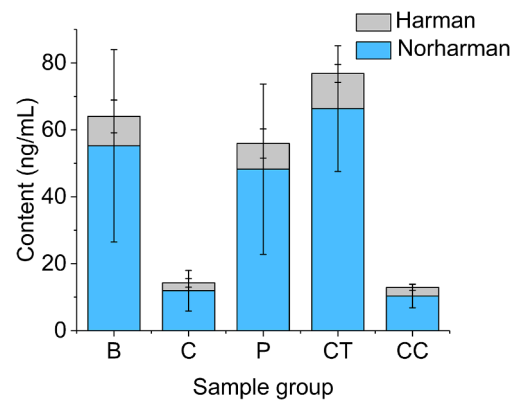

Fig. 5. The content of norharman and harman in experimental and commercial samples (B, boiled; C, cold-brew; P, pouring over; CT, commercial thermal-processed sample; CC, commercial cold-brew sample).

process, the temperature in brewing process, may play a role in increasing extraction of norharman and harman. Norharman and harman were also quantified in the 14 commercial coffee drink samples (Fig. 5, 7 thermally-processed samples and 7 cold-brew samples). The content difference of norharman and harman in commercial coffee samples were in highly consistent with the concentration changes measured in experimental samples.

4. Conclusions

An untargeted metabolomic approach based on UPLC-QTOF combining with chemometrics analysis provides an effective approach for differentiating coffee brewed by boiled, pour-over, and cold-brew methods, PCA of raw LC/MS data in both positive and negative ion modes could separate the three brewed methods in score plot. OPLS-DA model was built to filter the markers, and nine potential markers were putatively annotated by matching with online chemical databases. Additionally, five of them were further confirmed by UPLC-QQQ with the reference standards as characteristic markers. To the best of our knowledge, this is the first report of 1-palmitoyllysophosphatidylcholine and 4-hydroxy-3-methoxycinnamaldehyde in coffee drinks. Moreover, pyrimethanil is a cultivating addition, also methodology refinement on the development and validation of a target quantitative methods for 1-palmitoyllysophosphatidylcholine and 4-hydroxy-3-methoxycinnamaldehyde using UPLC-QQQ were still needed, therefore, only norharman and harman were selected for further verification and quantification in both experimental and commercial coffee samples. The two selected markers, harman and norharman, which present in

lower content in cold-brew samples than thermally processed samples (boiled and pour-over), were verified in commercial samples as well. Information about the health effect of these characteristic compounds, the change mechanisms of them under different brewed method, as well as their implications in the consumer decisions shall be studied in a multidisciplinary work. Last but not least, every brewing method has a particular and necessary grinding level of beans, powder/water ratio, etc., therefore, the extended numbers by adding commercial samples brewed by different methods are recommendable for the following study.

Conflict of interest

The authors declare no conflict of interest.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodchem.2019.01.154.

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